

Root Cause Analysis Report

Root Cause Analysis (RCA) Report – Account Compromise

1. Compromised Account Details:

- **Compromised System:** kkeshav/gm
- **Compromised Date:** 07-10-2025T01:47 PM
- **Compromised User:** K Keshav
- **Compromised Email Address:** kkeshav@grouppharma.in
- **Email Account Accessed Date:** 28-10-2025T02:51 PM
- **System:** System Compromise, Microsoft 365 (outlook)
- **Administrative Enabled:** Yes
- **MFA:** Enabled
- **Method of Compromise:** System Compromise + MITM + Credential Theft + Session Hijack + Typo squatting
- **Reported Time:** 2:18 PM, 05-11-2025

2. Summary:

This document provides a Root Cause Analysis (RCA) for a recent system compromise incident. The compromise occurred on October 07, 2025, enabling the attacker to obtain authentication data and subsequently gain unauthorized access to the account. The attacker-maintained persistence by reusing a valid authentication token, effectively bypassing MFA protection.

3. Timeline of Events:

Event Description	Date and Time (IST)	Source	Ipaddress	Location
The system was compromised.	Oct 7, 2025, 01:47 PM	Forensics tools		
The malware made external malicious requests.	Oct 7, 2025, 01:47 PM	Forensics tools		
The malware accessed the browser database files.	Oct 7, 2025, 07:24 PM	Forensics tools		
First Successful attacker login after	October 28, 2025 2:51 PM	Email/ Signin logs	102.89.76.52	Ibadan, Oyo State, Nigeria
The attacker created mailbox rules to automatically forward incoming email to an RSS feed in order to avoid detection.	October 28, 2025 2:51 PM	Email	102.89.76.52	Ibadan, Oyo State, Nigeria
A typosquatted domain named groupphrama.in was created by the attacker and used to impersonate Mahesh Babu.	Oct 28, 2025 07:33:47 AM	Email	102.89.76.52	Ibadan, Oyo State, Nigeria
The attacker exchanged a few emails and convinced the target to make the payment	Oct 28 to 04 Nov, 2025	Email	102.89.76.52	Ibadan, Oyo State, Nigeria
The payment was initiated by the client, as mentioned in the payment r	Nov 4, 2025 6:38 pm	Email	102.89.76.52	Ibadan, Oyo State, Nigeria

4. Indicators of Compromise:

- **Initial Indicator:** A malicious software was executed, Emails sent without user knowledge
- **Malicious File Name :** pw-free-online.exe
- **File Hash (MD5) :** 863B9F45B230138E828048B0A20370C7
- **Child Process Hash (Zero-day):**
757a832effed0f65f7a401d88b664f07f53dc53fd964f3c06dbbafd9c2aeba28
- **Sender Name:** k keshav
- **Sender Email:** kkeshav@grouppharma.in
- **Attacker Typo squatted Email in CC:** maheshbabu@groupphrama.in
- **Recipient:** naidukl@drreddys.com, ayyappak@drreddys.com, sirishap@drreddys.com, gashok@drreddys.com, MruduLavanyaPyna@drreddys.com, rohanj@drreddys.com, gauthamt@drreddys.com, psravyasree@drreddys.com, rammohand@drreddys.com, johnvkr@drreddys.com, rohanj@drreddys.com
- **Subject:** Re: [External]RE: DRL Invoice Status - 541255
- **IP Address:** 102.89.76.52
- **Unusual login location:** Yes (Ibadan, Oyo State, Nigeria).

5. Root Cause:

The credential, browser database, network traffic, cookies, and MFA theft occurred due to a **system compromise** during the installation of **MiniTool Partition Wizard**. Evidence indicates that the compromise was triggered by malicious components included with the installer, which executed unauthorized background processes and deployed hidden payloads that enabled credential harvesting.

A. System Compromise (Oct 7, 2025, 01:47 PM):

Forensic analysis using Amcache, JLECmd, and other artifact-review tools identified unauthorized process execution and abnormal activity, confirming a system-level compromise at this timestamp.

07-10-2025 13:47	c:\programdata\lenovo\systemupdate\sessionse\rep\fhbyd1042_1_version.exe	.exe		2707672
07-10-2025 13:47	c:\program files\pchealthcheck\pchealthcheckbroker PCHealthCheckBroker.exe	.exe		179120
07-10-2025 13:47	c:\program files\pchealthcheck\pchealthcheck.exe PCHealthCheck.exe	.exe		866224
07-10-2025 13:47	c:\users\gm\downloads\pw-free-online.exe pw-free-online.exe	.exe	minitool partition wizard	3254696
07-10-2025 13:47	c:\users\gm\appdata\local\synologydrive\synologydr cloud-drive-connect.exe	.exe	synology drive client	653808
07-10-2025 13:47	c:\users\gm\appdata\local\synologydrive\synologydr cloud-drive-daemon.exe	.exe	synology drive client	3751408
07-10-2025 13:47	c:\users\gm\appdata\local\synologydrive\synologydr cloud-drive-ui.exe	.exe	synology drive client	7377904
07-10-2025 13:47	c:\users\gm\appdata\local\synologydrive\synologydr cloud-drive-auto-updater.exe	.exe	synology drive client	1471472
07-10-2025 13:47	c:\programdata\lenovo\systemupdate\sessionse\rep\CHK.exe	.exe		308216

B. External Malicious Requests (Oct 7, 2025, 01:47 PM):

Network traffic captured and reviewed in Wireshark and supporting network-forensics tools showed the malware initiating outbound malicious connections immediately following the compromise.

No.	Time	Source	Destination	Protocol	Length	Info
6407	75.132434	10.0.14.131	172.217.174.78	TCP	66	[TCP Dup ACK 71#1] 36816 → 443 [ACK] Seq=1 Ack=1 Win=396 L
6408	75.132446	10.0.14.131	142.250.77.42	TCP	66	[TCP Dup ACK 72#1] 47248 → 443 [ACK] Seq=1 Ack=1 Win=670 L
6409	75.132452	10.0.14.131	151.101.65.229	TCP	66	[TCP Dup ACK 70#1] 54782 → 443 [ACK] Seq=1 Ack=1 Win=654 L
6410	75.133213	151.101.65.229	10.0.14.131	TCP	66	[TCP Dup ACK 73#1] [TCP ACKed unseen segment] 443 → 54782
6411	75.133953	172.217.174.78	10.0.14.131	TCP	66	[TCP Dup ACK 74#1] [TCP ACKed unseen segment] 443 → 36816
6412	75.134209	142.250.77.42	10.0.14.131	TCP	66	[TCP Dup ACK 75#1] [TCP ACKed unseen segment] 443 → 47248
6413	77.147368	87.120.191.104	10.0.14.131	TCP	60	43508 → 8728 [SYN] Seq=0 Win=65535 Len=0
6414	77.147453	10.0.14.131	87.120.191.104	TCP	54	8728 → 43508 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
6415	77.483491	10.0.14.131	8.8.8.8	DNS	76	Standard query 0x2a11 A www.minitool.com
6416	77.483504	10.0.14.131	8.8.8.8	DNS	76	Standard query 0xe512 AAAA www.minitool.com
6417	77.490859	8.8.8.8	10.0.14.131	DNS	108	Standard query response 0x2a11 A www.minitool.com A 104.18
6418	77.491682	8.8.8.8	10.0.14.131	DNS	132	Standard query response 0xe512 AAAA www.minitool.com AAAA
6419	77.492864	10.0.14.131	104.18.21.178	TCP	74	32960 → 443 [SYN] Seq=0 Win=62727 Len=0 MSS=8961 SACK_PERM

10
/95

Community Score -3

10/95 security vendors flagged this IP address as malicious

87.120.191.104 (87.120.191.0/24)

AS 215925 (Vpsvault.host Ltd)

DETECTION

DETAILS

RELATIONS

COMMUNITY 7

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

alphaMountain.ai

Malicious

ArcSight Threat Intelligence

Phishing

CRDF

Malicious

Cyble

Malicious

C. Browser Database Access (Oct 7, 2025, 07:24 PM):

Execution logs and additional forensic artifacts confirmed that the malware later accessed and interacted with browser database files, indicating attempts to extract stored credentials, session data, and authentication information.

settings.db	2025-10-07 19:26:55 IST
EdgeHubAppUsageSQLite.db	2025-10-07 19:26:37 IST
EdgeEDropSQLite.db	2025-10-07 19:25:18 IST
EdgeHubAppUsageSQLite.db	2025-10-07 19:25:18 IST
heavy_ad_intervention_opt_out.db	2025-10-07 19:25:16 IST
MediaDb.v1.sqlite	2025-10-07 19:25:16 IST
shared.sqlite	2025-10-07 19:25:16 IST
standalone.sqlite	2025-10-07 19:25:16 IST
plum.sqlite	2025-10-07 19:24:58 IST
EdgeEDropSQLite.db	2025-10-07 19:24:53 IST
EdgeHubAppUsageSQLite.db	2025-10-07 19:24:53 IST
heavy_ad_intervention_opt_out.db	2025-10-07 19:24:52 IST
fonts.db	2025-10-07 19:24:51 IST
AsyncCache.db	2025-10-07 19:24:32 IST
installed.db	2025-10-07 19:24:31 IST
installed.db	2025-10-07 19:24:31 IST
pinning.db	2025-10-07 19:24:31 IST
EdgeEDropSQLite.db	2025-10-07 19:24:24 IST
EdgeHubAppUsageSQLite.db	2025-10-07 19:24:24 IST
heavy_ad_intervention_opt_out.db	2025-10-07 19:24:22 IST

D. Attacker Observation Period (Oct 07 – Oct 27, 2025)

The attacker utilized this time period for further analysis and observation of the user and their activities.

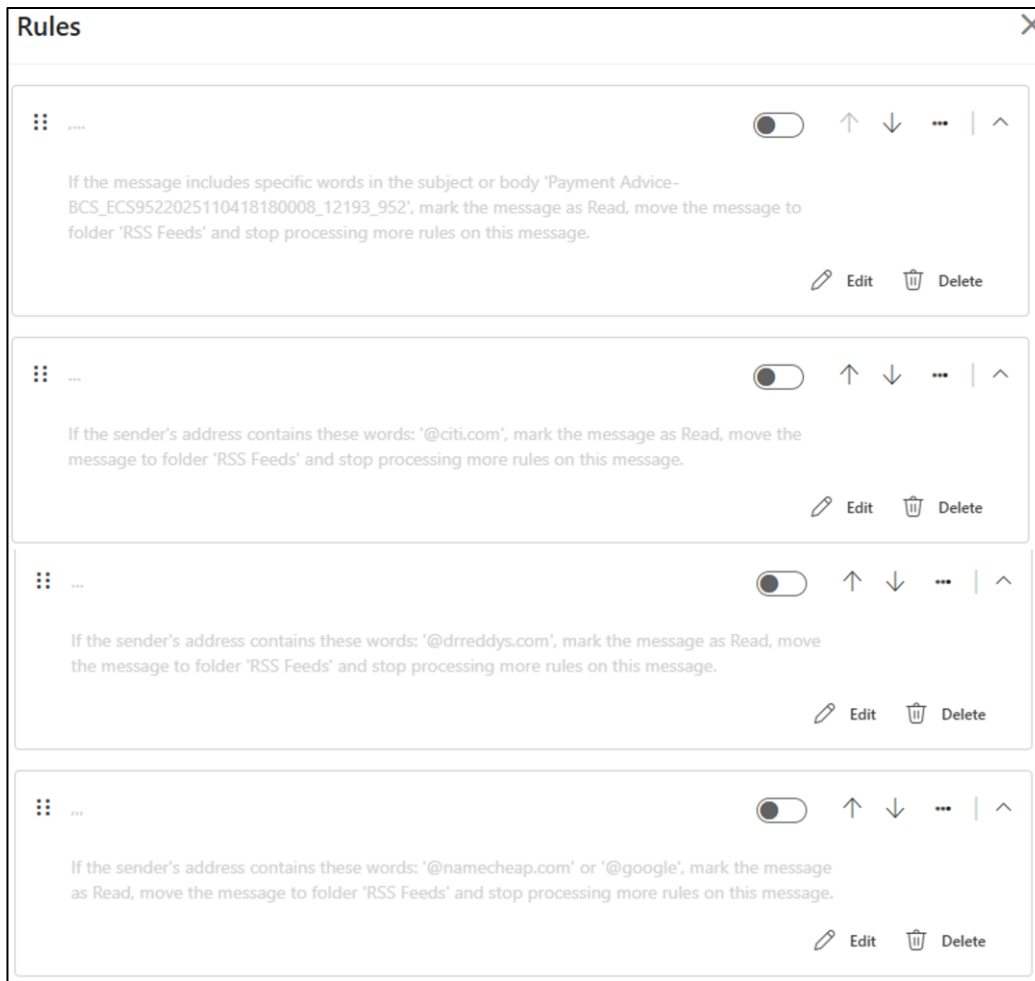
E. First Successful Attacker Login (October 28, 2025 – 2:51 PM)

The attacker successfully logged into the compromised account for the first time. This activity was confirmed through sign-in and email logs, marking the beginning of unauthorized access.

Date (UTC)	2025-10-28T10:58:49Z
Username	kkeshav@grouppharma.in
Application	One Outlook Web
Multifactor authentication result	MFA requirement satisfied by claim in the token
IP address	102.89.76.52
Location	,, NG

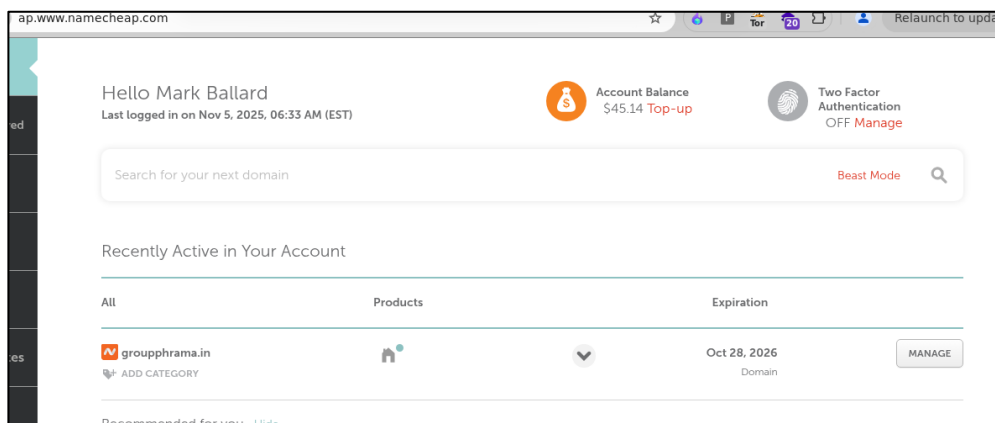
F. Mailbox Rule Creation (October 28, 2025 – 2:51 PM)

The attacker created mailbox rules to automatically forward incoming emails to an RSS feed. This action was intended to hide incoming messages and avoid detection by the legitimate user.



G. Typosquatted Domain Setup (October 28, 2025 – 07:33:47 AM)

A typosquatted domain, **grouphrama.in**, was registered by the attacker and used to impersonate Mahesh Babu. This domain was likely used to make fraudulent communications appear legitimate.



H. Social Engineering and Payment Manipulation (October 28 – November 4, 2025)

The attacker exchanged multiple emails with the target, building trust and eventually convincing the target to proceed with a payment request.

Re: [External]RE: DRL Invoice Status - 541255.

From: Keshav K <kkeshav@grouppharma.in>
Sent: 28 October 2025 18:17
To: Dr. T V Gautham <gauthamt@drreddys.com>; R John Vinay Kumar <johnvkr@drreddys.com>
Cc: D Ram Mohan <rammohand@drreddys.com>; KL Naidu <naidukl@drreddys.com>; G Ashok <gashok@drreddys.com>; Mrudu Lavanya Pyna <MrudulavanyaPyna@drreddys.com>; Mahesh Babu <maheshbabu@grouppharma.in>
Subject: Re: [External]RE: DRL Invoice Status - 541255.

Dear Dr Gautham,

Thank you,

In an effort to make payment remittance as seamless as possible also note that we have enhanced our accounting information

I. Payment Initiation (November 4, 2025 – 6:38 PM)

The client initiated the payment as indicated in the payment receipt, which was the attacker's fraudulent objective.

Payment Advice-BCS_I [REDACTED]

CAUTION: This email originated from outside of the organization. Do not click on links or open attachments unless you know the content is safe. Malware/ Viruses can be easily transmitted via email.

Payment Advice

For the attention of:

GROUP PHARMACEUTICALS LIMITED

6. Impact:

The system compromise enabled the attacker to extract authentication data, including a valid refresh token, which was subsequently used to gain unauthorized access to the user's account. Using this token-based access, the attacker viewed payment-related emails and created unauthorized mail-redirection rules, all without triggering standard login prompts or MFA challenges.

7. Actions Taken:

- The system has been isolated.
- Malicious processes have been terminated.
- A comprehensive malware analysis, forensic investigation, and reverse engineering assessment have been completed.
- The password has been reset
- All tokens & sessions were revoked, and re-authentication was enforced for the affected account.
- All email rules created by the attacker have been disabled.
- The typo squatted domain was accessed, and all DNS records were removed.

8. Key Takeaways:

1. The user's system had administrative-level access, which should be revoked or restricted.
2. Application whitelisting should be implemented to prevent unauthorized software execution.
3. Regular system and antivirus updates must be enforced.
4. Downloading or executing open-source applications from unknown or unverified third-party sources should be avoided.
5. Prior approval should be obtained before installing any unauthorized or non-standard applications.

9. Conclusion

The attacker gained access to the user's account by harvesting a valid refresh token following the system compromise. This allowed unauthorized access to emails and the creation of malicious mailbox rules to conceal activity. The issue has been contained by resetting the password, revoking active tokens, disabling unauthorized rules, and removing the typo squatted domain. Continuous monitoring and enhanced credential protection are recommended to prevent recurrence.